

REVIEW

Decoding Long Noncoding RNAs in Cardiac Development and Congenital Heart Disease: Recent Advances and Clinical Perspectives

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ABSTRACT: Long noncoding RNAs (lncRNAs) have emerged as crucial regulators in cardiac development and congenital heart disease. With advances in high-throughput sequencing, single-cell omics, and epigenetic profiling, accumulating evidence suggests that lncRNAs participate in transcriptional, posttranscriptional, and structural remodeling processes underlying cardiac morphogenesis. This in-depth review summarizes recent progress in elucidating the roles of lncRNAs in heart development and congenital heart disease pathogenesis. Specifically, we discuss how lncRNAs modulate key developmental signaling pathways, including Notch, Wnt/ β -catenin, BMP, and TGF- β , through interactions with chromatin modifiers, transcription factors, and microRNAs. Representative cardiac-enriched lncRNAs, such as *Braveheart*, *Fendrr*, *CARMEN*, *H19*, *MALAT1*, and *TBX5-AS1*, are highlighted for their roles in cardiomyocyte lineage commitment, endothelial-to-mesenchymal transition, endocardial–myocardial interactions, and structural morphogenesis. In various congenital heart disease subtypes, including ventricular septal defect, atrial septal defect, tetralogy of Fallot, and hypoplastic left heart syndrome, lncRNAs exhibit disease-specific expression patterns and participate in regulatory networks involving alternative splicing, competing endogenous RNA interactions, and genomic structural variations. Moreover, circulating and exosomal lncRNAs detected in maternal plasma and amniotic fluid have shown promise as candidate biomarkers for prenatal diagnoses. Collectively, these findings reveal that lncRNAs function as multilayered regulators, linking genetic, epigenetic, and environmental factors during cardiac development. A deeper understanding of lncRNA-mediated mechanisms may provide novel molecular targets, biomarkers, and diagnostic strategies for precision prevention and management of newborn infants with congenital heart disease.

Key Words: biomarkers ■ infant ■ morbidity ■ morphogenesis ■ plasma

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Congenital heart disease (CHD) is one of the most common congenital malformations, accounting for approximately one-third of all neonatal birth defects worldwide and representing the leading cause of morbidity and death during infancy. CHD prevalence varies significantly across regions, influenced by differences in diagnostic capacity and public health resources. According to a global study, the incidence rate of CHD is estimated to be 8 to 10 per 1000 live births, with detection rates in East Asia and parts of sub-Saharan Africa continuing to rise in recent years, mainly due to the widespread application of fetal echocardiography and neonatal screening technologies.¹

With advances in prenatal diagnosis and surgical interventions, the survival rate of patients with CHD

has increased markedly; however, long-term complications and cardiac dysfunction remain major challenges in both pediatric and adult cardiology.² CHD can be classified according to its anatomic and hemodynamic characteristics into shunt lesions such as ventricular septal defect (VSD) and atrial septal defect (ASD), obstructive lesions such as coarctation of the aorta and pulmonary atresia, and complex lesions such as tetralogy of Fallot (TOF) and hypoplastic left heart syndrome (HLHS).³

From the perspectives of cardiac developmental biology and pathology, CHD results from aberrant morphogenesis during early embryogenesis, involving key processes such as cardiac looping, chamber septation, valve morphogenesis, and outflow tract remodeling.

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Nonstandard Abbreviations and Acronyms	
ASD	atrial septal defect
ceRNA	competing endogenous RNA (ceRNA)
CHD	congenital heart disease
CNV	copy number variant
hiPSC	human induced pluripotent stem cell
HLHS	hypoplastic left heart syndrome
LncRNA	long noncoding RNA
PRC2	polycomb repressive complex 2
TOF	tetralogy of Fallot
TrxG	trithorax group
VSD	ventricular septal defect

During heart development, signaling pathways, including Notch, Wnt, BMP, and TGF- β , together with cardiac-specific transcription factors, such as NKX2-5, GATA4, and TBX5, synergistically regulate cardiomyocyte differentiation and morphogenesis. Disruption of these regulatory mechanisms can lead to a variety of structural malformations.⁴ Recent genomic studies have revealed that CHD exhibits both prominent polygenic and Mendelian inheritance patterns. An analysis of 5424 affected children revealed that recessive mutations account for $\approx$ 3% to 5% of pathogenic variants, suggesting a complex genetic contribution.⁵

Evidence-based evaluations suggest that only a minority of CHD cases can be explained by currently established monogenic causes,⁶ underscoring that most of the CHD cases arise from multifactorial interactions between genetic susceptibility and environmental influences. In addition to epigenetic mechanisms, such as DNA methylation and noncoding RNA regulation, accumulating evidence highlights the contribution of maternal environmental exposures and preexisting conditions. Maternal exposure to environmental pollutants, industrial chemicals, and teratogenic agents, as well as metabolic disorders including diabetes and obesity, have been associated with an increased risk of CHD in offspring. These nongenetic factors may interact with genetic predisposition to disrupt cardiac developmental pathways during cardiogenesis, thereby contributing to CHD pathogenesis.^{7–9}

At the diagnostic level, the detection of CHD is shifting from traditional morphological phenotyping to molecular and multiomics approaches. The integration of genomic next-generation sequencing with epigenomic studies and single-cell sequencing technologies has not only improved the detection rate of pathogenic variants but also facilitated establishing the causal relationship between specific mutations and clinical phenotypes, thereby providing a theoretical foundation for early diagnosis, risk prediction, and personalized intervention.¹⁰

In summary, the epidemiological characteristics of CHD exhibit marked regional variation worldwide, while its pathological basis lies in the complex imbalance of embryonic cardiac developmental networks and their intricate genetic/epigenetic regulatory systems.

Definition and Biological Functions of LncRNAs

Long noncoding RNAs (LncRNAs) are RNA molecules longer than 200 nucleotides that generally lack protein-coding potential.¹¹ With the advancement of high-throughput sequencing and transcriptomic technologies, lncRNAs have been found to be widely transcribed across eukaryotic genomes, outnumbering protein-coding genes, and located within exonic and intronic regions, antisense strands, and distal regulatory elements.^{12,13} According to their genomic position and relationship with neighboring genes, lncRNAs can be classified into several categories, including sense, antisense, bidirectional, intronic, intergenic, and enhancer-associated subtypes.^{14,15}

From a molecular perspective, lncRNAs participate in regulating gene expression through multilayered mechanisms, including epigenetic modification, transcriptional control, and posttranscriptional regulation (Figure 1). These regulatory functions are highly dependent on subcellular localization. In the nucleus, lncRNAs often function as guides or scaffolds to recruit chromatin-modifying complexes, such as PRC2 (polycomb repressive complex 2) and TrxG (trithorax group)/MLL complexes, thereby regulating gene expression in cis or trans and modulating higher-order chromatin organization, including 3-dimensional genome architecture and topologically associating domains.^{12,16} In the cytoplasm, lncRNAs contribute to posttranscriptional regulation by influencing mRNA stability and translational efficiency. A well-characterized mechanism involves their role as competing endogenous RNAs (ceRNAs), in which lncRNAs act as molecular sponges for microRNAs.¹⁷ Across cellular compartments, lncRNAs can also interact with proteins, acting as molecular scaffolds or decoys, thereby modulating the activity of transcription factors, RNA-binding proteins, and chromatin-modifying enzymes.^{12,16} Collectively, these findings highlight that lncRNAs exert context-dependent regulatory functions through coordinated nuclear and cytoplasmic mechanisms, RNA–protein interactions, and spatially controlled subcellular localization.^{12–17}

In recent years, lncRNAs have also been identified as mediators of mechanotransduction by regulating the Hippo–YAP signaling pathway, thereby participating in cell growth, migration, and tissue development.¹⁸ These diverse mechanisms collectively underscore the context-dependent and spatially regulated functions of lncRNAs. With the rapid advancement of bioinformatics, artificial intelligence, and systems biology, the roles of lncRNAs within disease networks have been further

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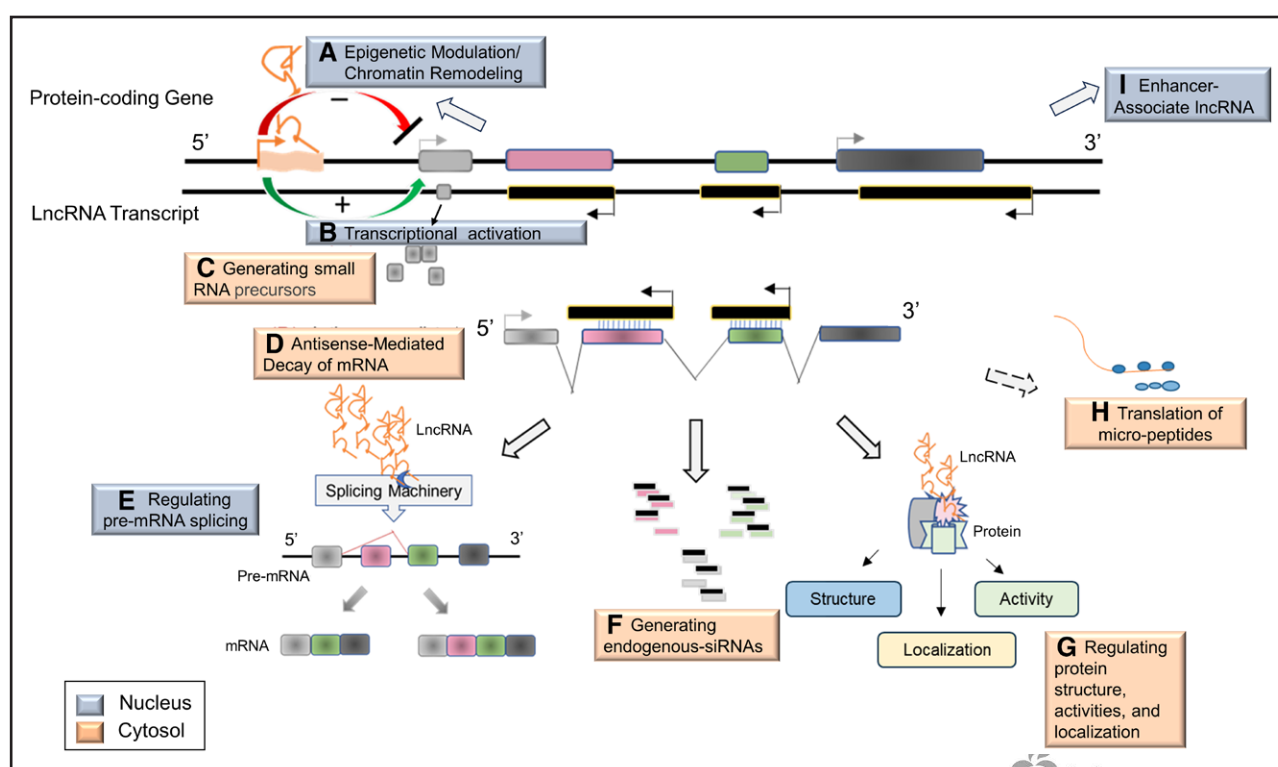


Figure 1. Biological functions and regulatory mechanisms of long noncoding RNAs (LncRNAs) relevant to subcellular localization.

Schematic summary of LncRNAs mechanisms of function (A and B) modulating chromatin remodeling and transcriptional activity. C, Serving as precursors to small RNAs. D, Acting as antisense transcripts that directly bind mRNA. E, Altering RNA splicing and processing events. F, Generating competitive endogenous RNAs (CeRNAs). G, Regulating protein structure, localization, and activity. H, Translation of micro-peptides. I, Enhancer-Associated lncRNAs. Subcellular compartments of LncRNAs functional mechanisms are highlighted. Blue box (nucleus) or orange box (cytosol).

elucidated. Analyses based on heterogeneous network models and graph attention auto-encoders have enabled effective extraction of nonlinear representations and improved prediction of potential lncRNA-disease associations.¹⁹ Transcriptome-scale CRISPR screening experiments have further validated the critical roles of specific lncRNAs in maintaining cell survival, hemostasis, and differentiation.²⁰

It is noteworthy that although lncRNAs are typically defined as noncoding transcripts, translatome-based studies have demonstrated that a subset of these molecules contains short open reading frames capable of being translated into functional micro-peptides. These micro-peptides engage in diverse biological processes, including energy metabolism, myocardial contraction, and signal transduction. Translatome analyses of the human heart have further revealed that a substantial number of lncRNAs exhibit potential translational activity, suggesting that their functions may extend far beyond the conventional definition of noncoding.²¹

In summary, lncRNAs represent multifunctional key players within the gene regulatory network, participating not only in epigenetic and transcriptional regulation but also in mediating cellular physiological processes and molecular properties through the interaction with DNA,

RNA, and protein molecules and the potential translation of functional micro-peptides (Figure 1). These findings provide a new perspective for understanding the molecular bases of their functions in cardiac development and disease.

LncRNAs in Cardiac Development

Cardiac development is a complex biological process that requires precise coordination among multiple cellular lineages and signaling pathways. In early embryogenesis, the heart originates from the cardiac mesoderm, which subsequently undergoes heart tube formation, ventricular formation, and septation, ultimately giving rise to the 4-chambered heart.²² During this process, several key signaling pathways—including Wnt/ β -catenin, BMP, Notch, TGF- β , and FGF cascades—cooperatively regulate the fate determination, proliferation, and differentiation of cardiac progenitor cells.^{23–25} The spatial-temporal activation and inhibition of these signaling pathways are crucial for maintaining proper cardiac morphogenesis and the normal development of myocardial electrophysiological properties. Recent studies have revealed that lncRNAs exert fine regulatory control within these core pathways across several stages of cardiac development

by interacting with key cardiac transcription factors, chromatin remodeler complexes, and microRNA networks,¹¹ thereby establishing a multilayered regulatory framework that fine-tunes cardiac gene expression.²⁶ To provide a more comprehensive overview, lncRNA functions in cardiac development can be broadly categorized according to key developmental stages, including cardiac lineage commitment, cardiomyocyte differentiation and maturation, and chamber specification.

Key lncRNAs Regulating Cardiac Lineage Commitment

During cardiac lineage differentiation and structural development, several lncRNAs have been identified to play essential functional roles. The first reported cardiac-specific lncRNA, Braveheart (*Bvht*), was discovered in a mouse embryonic stem cell model. *Bvht* promotes the transcriptional activation of cardiac lineage-specific genes, such as *Mesp1* and *Nkx2-5*, by interacting with the SUZ12/PRC2 complex.²⁷ Knockout of *Bvht* results in a blockade of cardiomyocyte differentiation from embryonic stem cells, indicating that it serves as a core noncoding regulatory factor essential for cardiac lineage establishment. *FENDRR* is a lncRNA highly expressed during early mesoderm and cardiac development. By interacting with the PRC2 and the TrxG/Mll complexes, *FENDRR* regulates the histone methylation status of developmental genes such as *Foxf1* and *Pitx2*, thereby maintaining left-right ventricular symmetry and proper pericardial septation during heart morphogenesis.²⁸ *Upperhand (Uph)* is coexpressed with its neighboring gene *Hand2* and participates in its cis-regulatory control. Loss of *Uph* results in abnormal cardiomyocyte differentiation and right ventricular developmental defects.²⁹ *CARMEN* ([C]ardiac [M]esoderm [E]nhancer-associated [N]oncoding RNA) regulates the balance between myocardial and smooth muscle lineages in cardiovascular progenitor cells by interacting with SMAD2/3 signaling, serving as a critical transcriptional link between cardiac and vascular development.^{30,31}

Recent studies have further expanded the repertoire of lncRNAs involved in early cardiac lineage commitment. For example, the Linc1405–Eomes complex has been shown to promote cardiac mesoderm specification in both mouse embryonic stem cells and in vivo models, highlighting the importance of lncRNA-mediated regulation at the earliest stages of cardiogenesis.³² In addition, lncRNAs such as H19 and MALAT1 have been shown to exert bidirectional regulatory effects on embryonic cardiomyocyte migration and heart tube formation. These effects are partly mediated through key developmental signaling pathways, including Wnt/ β -catenin, Notch, and TGF- β signaling.^{33–35}

Beyond early lineage commitment, lncRNAs also play essential roles in later stages of cardiac development.

HBL1 has been identified as a human-specific lncRNA that suppresses cardiomyocyte differentiation from human induced pluripotent stem cells (hiPSCs) by antagonizing miR-1 activity,³⁶ whereas CAREL regulates postnatal cardiomyocyte cell cycle arrest and cardiac maturation.³⁷ In addition, chamber-specific expression patterns of lncRNAs have been observed between the left and right ventricles, suggesting roles in chamber specification and morphogenesis. *Ppp1r1b-lncRNA/Tcap* axis regulates sarcomere gene expression during perinatal cardiac maturation. Further, the expression ratio differentiates TOF from VSD in human infantile hearts.³⁸ Imprinted lncRNAs such as *Kcnq1ot1* further regulate chromatin architecture and gene expression at disease-associated loci.³⁹

Cell Type-Specific Roles of lncRNAs in Cardiac Development

lncRNAs exhibit pronounced cell type-specificity in their expression and function. Previous studies have demonstrated that cell populations residing in distinct anatomic regions and developmental stages of the heart exhibit unique lncRNA expression profiles.⁴⁰ Importantly, accumulating evidence suggests that many lncRNAs involved in cardiac development have also been implicated in adult cardiovascular diseases, indicating that fetal lncRNA-mediated regulatory programs may persist or re-emerge under pathological conditions.⁴¹

In cardiomyocytes, lncRNAs such as myosin heavy chain associated RNA transcript have been shown to regulate stress responses and pathological remodeling. Myosin heavy chain associated RNA transcript interacts with the chromatin remodeler BRG1 to inhibit aberrant gene activation, thereby protecting the heart from pathological hypertrophy.⁴² Similarly, lncRNAs such as MALAT1, which is discussed in the context of cardiovascular disease in later sections, have also been implicated in regulating cardiomyocyte proliferation, apoptosis, and stress adaptation, suggesting that their functions may originate from or be closely linked to developmental regulatory pathways.^{41,43}

In endocardial and endocardial cushion cells, H19 has been implicated in the regulation of endothelial-to-mesenchymal transition and cardiovascular developmental processes.³³ Notably, dysregulation of endothelial-to-mesenchymal transition-related pathways has been associated with valvular diseases and cardiac fibrosis in adulthood,^{35,44} further supporting the concept that developmental regulatory programs, potentially involving lncRNA-mediated mechanisms, contribute to disease susceptibility.^{45,46}

In cardiac fibroblasts, *Wisper* has been found to promote fibroblast activation and cardiac fibrosis, and its expression is markedly upregulated following cardiac injury.⁴⁷ This observation highlights a potential link

between fibroblast-associated lncRNA regulation and pathological remodeling processes in the adult heart.⁴⁷ consistent with the broader concept that developmental lncRNA programs may be reactivated in disease contexts.^{41,47}

In the vascular smooth muscle cell lineage, lncRNAs such as smooth muscle enriched lncRNA have been implicated in regulating smooth muscle cell proliferation and phenotypic modulation, thereby contributing to vascular remodeling processes.⁴⁸ Alterations in these regulatory pathways may contribute to vascular remodeling and congenital and acquired cardiovascular diseases.^{41,48}

Overall, the roles of lncRNAs in cardiac development have expanded from single-gene regulation to cross-lineage molecular cellular regulatory networks. By coordinating chromatin modification, transcriptional regulation, and signaling pathway activation, lncRNAs achieve precise control over cardiac lineage formation, tissue/cell specificity, and structural morphogenesis (Table 1), thereby providing a molecular basis for understanding congenital cardiac structural abnormalities. Moreover, the reactivation or dysregulation of developmental lncRNA programs in adult cardiovascular diseases suggests that they may serve as mechanistic links between early cardiac development and later disease phenotypes.²⁶

LncRNAs in CHD

The pathogenesis of CHD is complex and multifactorial. In addition to classical structural gene mutations, dysregulation of the noncoding genome has recently been recognized as a crucial molecular basis of CHD. Recent transcriptomic and single-cell studies have further revealed that significant alterations in lncRNA expression profiles occur in cardiac tissues from patients with distinct types of CHD, including VSD, TOF, and HLHS. Notably, several of these lncRNAs exhibit strong coexpression relationships with genes involved in cardiac development and fetal gene program.^{40,46} (Figure 2).

LncRNAs may mediate key developmental processes by regulating Notch and BMP signaling pathways during endocardial–myocardial interactions.⁴⁶ Further investigations have demonstrated that aberrant RNA splicing represents another key mechanism by which lncRNAs contribute to CHD pathogenesis. In heterotaxy models, splicing abnormalities within noncoding regions lead to the generation of multiple abnormal transcript isoforms, thereby disrupting left–right axis signaling balance and resulting in complex cardiac developmental defects.⁴⁹ In addition to splicing abnormalities, genomic structural variations—such as copy number variants (CNVs)—can also affect the expression and function of lncRNAs. Studies integrating clinical CNV databases with lncRNA annotation data have revealed that certain lncRNA genes located within pathogenic CNV regions are closely associated with clinical phenotypes of CHD, suggesting that they may serve as candidate regulatory genes or putative pathogenic factors.⁵⁰

Importantly, the functions of lncRNAs are not confined to the heart itself, as their regulatory roles at the maternal–fetal interface and during placental development are also strongly associated with CHD risk. Emerging evidence indicates that noncoding RNAs participate in placental angiogenesis, immune tolerance, and nutrient transport, and that abnormal maternal–fetal lncRNA expression may indirectly contribute to CHD by disrupting fetal circulatory development. Based on this placenta–heart axis concept, lncRNAs are considered key molecular mediators linking the maternal metabolic environment with fetal cardiac development.⁵¹

In summary, lncRNAs participate in the occurrence and progression of CHD through several regulatory mechanisms, including transcriptional control, signaling networks, splicing machinery, epigenetic modulation, genomic structural variation, and maternal–fetal developmental pathways. Their disease-specific expression patterns and regulatory versatility make them a promising entry point for elucidating the noncoding genetic basis

Table 1. LncRNAs in Cardiac Development

LncRNA name	Function/mechanism (simplified)	Expression site/stage	Model system	Reference
<i>Braveheart (Bvht)</i>	Promotes cardiac lineage commitment	Embryonic mesoderm	Mouse embryonic stem cells	27
<i>FENDRR</i>	Regulates gene expression and heart symmetry	Early cardiac mesoderm	Mouse embryo	28
<i>Upperhand (Uph)</i>	Controls Hand2 expression and heart chamber development	Hand2-positive cardiac progenitors	Mouse	29
<i>CARMEN</i>	Balances muscle differentiation in heart progenitors	Cardiac progenitor cells	Human	30,31
<i>H19</i>	Supports cardiomyocyte migration and heart tube formation	Embryonic heart	Mouse	33
<i>MALAT1</i>	Guides cardiomyocyte migration and heart development	Embryonic heart	Mouse	34
<i>MHRT</i>	Prevents pathological heart remodeling	Adult cardiomyocytes	Mouse	42
<i>Wisper</i>	Promotes heart fibrosis	Cardiac fibroblasts	Mouse	47
<i>SMILR</i>	Maintains vascular muscle cell function	Vascular smooth muscle cells	Human	48

LncRNA indicates long noncoding RNA.

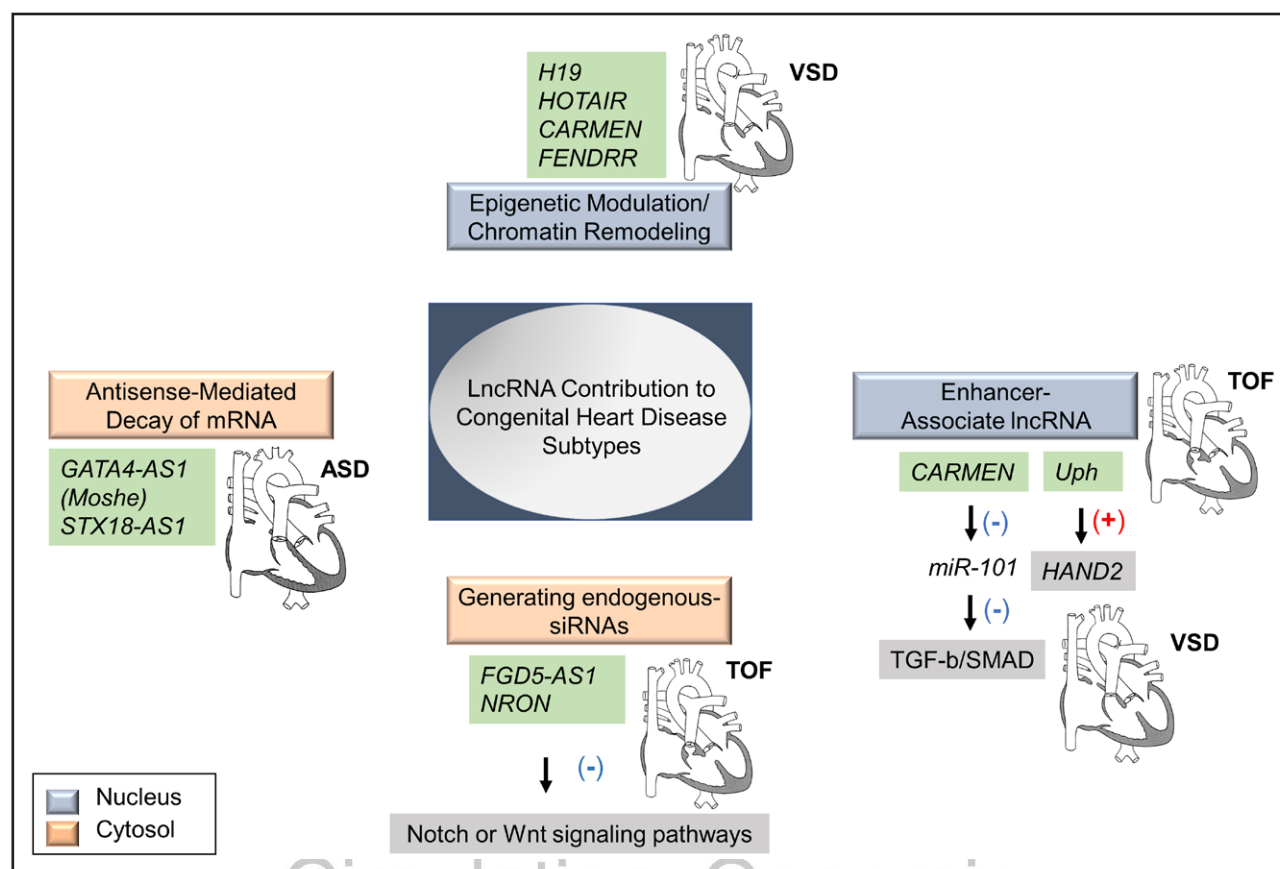


Figure 2. Long noncoding RNAs (lncRNAs) contribution to congenital heart disease (CHD) subtypes.

Schematic summary of example lncRNAs (green boxes) associated with CHD subtypes including atrial septal defect (ASD), ventricular septal defect (VSD), and tetralogy of Fallot (TOF). Functional mechanisms and subcellular compartments (nucleus vs cytosol) are highlighted. Blue box (nucleus) or orange box (cytosol).

of CHD and for developing novel molecular diagnostic and targeted lncRNA-based therapeutic strategies.⁵⁰ In the following sections, we summarize recent progress in elucidating the roles of lncRNAs in CHD subtypes.

lncRNAs in VSD

VSD is one of the most common types of CHD. Among all CHD subtypes, VSD accounts for $\approx 30\%$ to 40% of cases, with an estimated incidence rate of 4–6 per 1000 live births.⁵² Several studies have revealed significant dysregulation of lncRNA expression in fetal cardiac tissues and maternal body fluids from VSD cases, suggesting that these molecules may participate in ventricular septum formation by influencing cardiomyocyte differentiation, signal pathway activation, and intercellular communication.^{53,54}

At the scale of development, dysregulated expression of the imprinted lncRNA *H19/Igf2* locus has been identified as an important cause of cardiac-placental axis disruption. Its aberrant activation disturbs the metabolic and signaling balance between the embryonic heart and placenta, thereby leading to cardiac defects, including VSD, resembling Silver–Russell syndrome.⁵⁵ This finding

provides novel molecular evidence linking the maternal–fetal metabolic environment to cardiac malformations. Further investigations have revealed that *SNHG6* promotes aberrant activation of the Wnt/ β -catenin signaling pathway by negatively regulating *miR-101*, thereby releasing its inhibitory effect on downstream targets and ultimately impairing cardiac structural formation.⁵⁶ This study not only elucidates the critical regulatory role of the lncRNA–miRNA–mRNA network in cardiac development but also provides a novel molecular explanation for the pathogenesis of VSD. Additionally, bioinformatic analysis combined with experimental validation has identified lncRNA *TUC40-* as a molecule associated with VSD. This lncRNA is highly expressed during cardiomyocyte differentiation, and, in P19 cell–based cardiac differentiation models, it has been shown to promote cardiomyocyte differentiation and functional maturation.^{57,58} These findings suggest that *TUC40-* may promote myocardial lineage development during ventricular septum formation, providing a potential therapeutic target for early intervention.

With respect to signaling mechanisms, Notch signaling and its interactions with the Wnt and TGF- β pathways are considered key molecular mechanisms regulating

cardiac septum formation. The Notch pathway maintains structural symmetry and integrity of the developing heart by controlling the proliferation and differentiation of cardiac progenitor cells, and its dysregulation has been strongly associated with the development of VSD.²³ As stated above, it is noteworthy that lncRNAs exhibit complex interactions with these developmental signaling pathways, potentially amplifying signaling perturbations through upstream transcriptional regulation or competitive binding to downstream mediators. At the upstream level, certain lncRNAs (such as *H19* and *CARMEN*) can modulate chromatin accessibility and promoter activity of genes associated with the Wnt/ β -catenin or TGF- β /SMAD pathways, thereby enhancing signal responsiveness.²⁶ At the downstream level, lncRNAs can act as ceRNAs Sponge that sequester microRNAs (eg, miR-19, miR-101), thereby relieving their repression of key transcription factors in the Notch and Wnt signaling pathways. Through these multilevel mechanisms, lncRNAs can amplify signaling perturbations, leading to imbalances in cardiomyocyte lineage specification and septal morphogenesis.⁵⁹

In the context of genetic regulation, large-scale CNV analysis has revealed that a subset of lncRNAs located within structural variation hotspot regions may act as susceptibility genes or regulatory elements involved in the development of VSD.⁵⁰ Such structural genomic variations highlight the potential contribution of noncoding genes to CHD susceptibility, providing a new dimension for genetic diagnosis.

Integrated transcriptomic and systemic biology analyses have further revealed the presence of multiple dysregulated lncRNA–mRNA co-expression axes in fetal cardiac tissues from VSD cases.⁵³ These findings indicate the broad regulatory roles of lncRNAs across gene network systems. Importantly, a series of differentially expressed lncRNAs and ceRNA networks have also been identified in amniotic fluid and amniotic fluid-derived exosomes.^{54,60} These noncoding molecules may mediate maternal–fetal signaling interactions and participate in embryonic cardiac development. This discovery not only expands our understanding of the molecular mechanisms underlying VSD but also provides potential biomarkers for noninvasive prenatal detection in clinical practice.

Collectively, current evidence supports the multilayered regulatory mechanisms of lncRNAs in the development of VSD, including: (1) modulating embryonic cardiac morphogenesis through imprinting regulation and the placenta–heart developmental axis⁵⁵; (2) regulating cardiomyocyte differentiation and septum formation through key developmental pathways and lncRNA–miRNA–mRNA regulatory networks^{54,60,61}; (3) acting as regulatory factors within genomic structural variations, thereby contributing to the formation of genetic susceptibility.⁵⁰ In the context of translational regulation, lncRNAs, particularly

those encapsulated within exosomes, hold promise as novel biomarkers for early diagnosis and intervention of VSD in the future.⁶⁰

lncRNAs in ASD

ASD is one of the most common septal malformations in CHD. Its incidence ranks second only to VSD, accounting for $\approx 10\%$ to 15% of all CHD cases, with an estimated incidence rate of 1 to 2 per 1000 live births.⁶² High-throughput sequencing and functional validation studies have demonstrated that several lncRNAs exhibit spatiotemporal expression patterns during critical stages of heart development, and their dysregulation may disrupt cardiomyocyte differentiation, signaling pathway modulation, and lineage equilibrium, thereby contributing to the molecular pathogenesis of ASD.⁶²

lncRNA *Moshe*, one of the *Gata6* antisense transcripts (*GATA4-AS1*) located upstream of *Gata6*, has been identified as a heart-specific transcript that exhibits distinct spatiotemporal expression patterns during the second heart field stage. It cooperates with cardiac transcription factor networks, including *Nkx2-5* and *Gata4*, to regulate myocardial lineage commitment, thereby influencing the migration and organization of cardiac progenitor cells.⁶³ Aberrant expression of the murine ortholog of *GATA6-AS1*, *Moshe*, may lead to imbalanced lineage allocation during atrial septum formation, providing a novel mechanistic explanation for the pathogenesis of ASD.

Systematic analyses have revealed that noncoding RNAs (including miRNAs, lncRNAs, and circRNAs) form complex regulatory networks spanning multiple stages from cardiac tube formation to atrial septum closure. In particular, the lncRNA *STX18-AS1*, located on chromosome 4p16, has been identified in multiple genome-wide association studies as a major susceptibility gene for ASD.⁶⁴ Further in vitro experiments have demonstrated that *STX18-AS1* regulates the methylation status and expression rhythm of the key cardiac transcription factor *NKX2-5*, thereby influencing the direction and rate of cardiomyocyte progenitor differentiation and ultimately modulating the developmental progression of the atrial septum.^{65,66} These series of studies have, for the first time, established a nearly complete causal regulatory chain linking genetic variation–lncRNA regulation–cardiac development–structural defect, providing a solid foundation for understanding the noncoding regulatory mechanisms underlying ASD.

Meanwhile, studies on circulating lncRNAs have opened new avenues for the early detection and noninvasive screening of ASD. Analyses of maternal plasma and amniotic fluid samples during pregnancy have revealed that multiple lncRNAs are differentially expressed in fetuses with CHD, and these diagnostic signatures exhibit favorable stability and reproducibility.^{67,68} Certain

lncRNAs, such as *STX18-AS1*, can be detected in maternal circulation, suggesting their potential as biomarkers for ASD.⁵¹ Recent studies have suggested the existence of a lncRNA-mediated signaling axis between the heart and placenta. Exosomes can transport specific lncRNAs across the maternal–fetal barrier, thereby regulating embryonic cardiac energy metabolism and angiogenesis during development.^{55,68} This maternal–fetal communication mechanism reveals the interwoven roles of environmental factors and epigenetic regulation in the formation of ASD and provides a new direction for early prenatal molecular intervention.

In summary, lncRNAs contribute to the pathogenesis of ASD through 3 major mechanisms: (1) acting as genetic susceptibility mediators that link genomic variation to cardiac developmental signaling networks (eg, *STX18-AS1*); (2) functioning as developmental regulators that modulate lineage specification and atrial septum formation during the second heart field stage (eg, *GATA6-AS1*); and (3) serving as transferable signaling molecules (eg, *H19*, *STX18-AS1*) that participate in maternal–fetal communication and exhibit detectability in body fluids. LncRNA contributions to VSD and ASD are summarized in Table 2.

LncRNAs in TOF

TOF represents the most common complex cyanotic CHD, accounting for ≈7% to 10% of all CHD cases, with an estimated incidence rate of 0.3 to 0.5 per 1000 live births.⁵² Classical TOF involves several developmental abnormalities in heart structure, including overriding aorta, pulmonary stenosis, right ventricular hypertrophy, and VSD. In recent years, transcriptomic sequencing and

systems biology analyses have revealed that lncRNAs play crucial roles in epigenetic regulation and signaling modulation in TOF pathogenesis and progression.^{69–71}

Across molecular interaction systems, transcriptomic analyses of right ventricular specimens have uncovered a profound reconstruction of lncRNA and mRNA expression profiles in the myocardial tissues of TOF patients. These lncRNAs participate in metabolic reprogramming and adaptive remodeling of cardiomyocytes through the formation of lncRNA–miRNA–mRNA regulatory networks.⁶⁹ Further studies using weighted gene coexpression network analysis have identified several hub lncRNAs closely associated with TOF, suggesting that these molecules may influence the expression of cardiac structural genes by regulating alternative splicing and chromatin modification.⁷² *FGD5-AS1* has been identified as a key ceRNA in the myocardium of TOF patients, capable of coordinately regulating multiple cardiac developmental pathways through transcriptional and epigenetic mechanisms.⁷³ Another nuclear lncRNA, *NRON*, is significantly downregulated in TOF hearts. Its loss suppresses the SHP2/Wnt/β-catenin signaling pathway and impairs cardiomyocyte differentiation, thereby revealing the signaling vulnerability during embryonic development in TOF.⁷⁰ Also, *HA117* is highly expressed in the right ventricular outflow tract tissues of TOF patients and is closely associated with anatomic subtypes and postoperative cardiac function, suggesting its potential as a biomarker for disease stratification and prognostic prediction.⁷⁴

With regard to epigenetic regulatory landscapes, hypermethylation of *TBX5-AS1:2* leads to its transcriptional repression, resulting in reduced *TBX5* expression and suppressed cardiomyocyte proliferation. This mechanism provides direct epigenetic evidence for TOF

Table 2. LncRNAs in VSD and ASD

LncRNA name	Function/mechanism (simplified)	Expression site/stage	Model system	Defect type	Reference
CNV-associated lncRNAs	Linked to genetic susceptibility through structural variation regions	CNV-rich genomic regions	Human (CNV analysis)	VSD	50
<i>H19</i>	Affects heart–placental signaling, disrupts septum formation	Embryonic (fetal) heart and placenta	Mouse model	VSD	55
<i>SNHG6</i>	Promotes cardiomyocyte proliferation via Wnt/β-catenin signaling	Fetal cardiac tissue (VSD patient hearts)	Human (fetal heart samples)	VSD	56
<i>TUC40</i>	Enhances cardiomyocyte differentiation and maturation	Cardiomyocyte induction stage	Mouse (P19 cells)	VSD	57
Exosomal lncRNAs	Participate in maternal–fetal communication	Maternal plasma and amniotic fluid (exosomes)	Human (fluid biopsies)	VSD	60
<i>GATA6-AS1</i> (Moshe)	Directs atrial progenitor development	Second heart field (embryonic heart)	Mouse embryonic heart	ASD	63
<i>STX18-AS1</i>	Regulates NKX2-5 expression and atrial septum formation	Atrial septum (embryonic heart)	Human (in vitro differentiation)	ASD	65
Circulating lncRNAs	Act as noninvasive biomarkers for fetal CHD	Maternal plasma and amniotic fluid	Human clinical samples	ASD	67
Exosomal lncRNAs	Involved in maternal–fetal communication and heart metabolism	Placenta (exosomes, prenatal)	Mouse maternal-placental model	ASD	55,68

ASD indicates atrial septal defect; CHD, congenital heart disease; CNV, copy number variation; LncRNA, long noncoding RNA; and TOF, tetralogy of Fallot; VSD, ventricular septal defect.

pathogenesis.⁷¹ Meanwhile, transcriptomic analyses of fetal TOF hearts have revealed coordinated differential expression of multiple lncRNAs and mRNAs, suggesting that the molecular pathology of the disease is established as early as the embryonic stage.⁷⁵

Perinatal biofluid-based studies have provided new evidence supporting the translational potential of lncRNAs. Maternal plasma and amniotic fluid lncRNAs can reflect the fetal cardiac developmental status, and several molecules exhibit promising sensitivity and specificity in CHD subtypes.⁷⁶ In addition, small RNA profiling studies have uncovered sex-specific differences in TOF patients,⁶¹ suggesting that sex-related epigenetic regulation may influence disease susceptibility and clinical heterogeneity.

Overall, lncRNAs contribute to the pathogenesis of TOF through several key mechanisms (Table 3): (1) regulating cardiac developmental genes at transcriptional and epigenetic levels; (2) forming ceRNA networks with miRNAs, thereby modulating key developmental signaling pathways, including Wnt/ β -catenin and TGF- β . In addition to these mechanistic roles, Table 3 also summarizes lncRNAs with potential clinical relevance, including those detectable in biofluids, highlighting their value as noninvasive biomarkers for disease screening and prognostic evaluation.

LncRNAs in Complex CHD

In complex CHDs, such as HLHS, hypoplastic right ventricle, transposition of the great arteries, and valvular malformations, lncRNAs are considered a crucial molecular layer that connects spatiotemporal regulation, lineage fate determination, and organ morphogenesis.

Taking HLHS as an example, single-cell and iPSC-based studies revealed that, beyond cardiomyocyte differentiation defects, HLHS also involves intrinsic endocardial abnormalities, characterized by the downregulation of NOTCH signaling, extracellular matrix remodeling, and induction of the endothelial-to-mesenchymal transition pathways, accompanied by impaired fibroblast–integrin signaling, which together lead to valvular obstruction and left ventricular hypoplasia.⁴⁶ LncRNAs are proposed to bridge genetic mutations and

developmental abnormalities by modulating NOTCH, Wnt/ β -catenin, and TGF- β pathways and maintaining the homeostasis of the fetal heart–placenta axis, thereby influencing left heart formation.^{51,77} Although direct studies specifically addressing lncRNAs in HLHS are still lacking, these findings provide new mechanistic insights into its molecular pathogenesis.

During the formation of the second heart field and outflow tract—critical processes that determine great vessel alignment and positioning—several lncRNAs have been identified to exert causal-level regulatory effects. For instance, the enhancer-associated lncRNA *Upperhand* (*Uph*), located at the *HAND2* locus, is essential for maintaining *HAND2* super-enhancer activity and proper development of the right ventricle and outflow tract. Importantly, blocking the transcription of *Uph* rather than its RNA product results in right ventricular hypoplasia and embryonic lethality, underscoring the functional necessity of *Uph* in cardiac morphogenesis.²⁹ On the contrary, the lateral plate mesoderm-specific lncRNA *FENDRR* modulates PRC2 to influence cardiac and body wall morphogenesis, highlighting its central role in early lineage specification and cardiac tube formation.^{26,28}

During the stage of valvulogenesis, particularly during the formation of the cardiac neural crest and endocardial cushions, studies have identified lncRNA *GATA6-AS1* and its ortholog *Mohshe*, as key regulators of aortic valve development.⁷⁸ It exhibits spatiotemporally specific expression during embryogenesis, and its loss of function results in a bicuspid aortic valve-like phenotype, accompanied by downregulation of valve development genes associated with the NOTCH and TGF- β pathways, suggesting that lncRNAs may serve as upstream regulators of valvular malformations.⁷⁸

From a mechanistic and translational perspective, lncRNA networks integrate cardiac transcription factors (such as NKX2-5, GATA4/6, and *HAND2*) with developmental signaling pathways at both transcriptional and epigenetic levels. Through ceRNA-mediated mechanisms, they form competitive interactions with miRNAs, thereby reshaping cell fate determination and metabolic coupling. These regulatory principles have been validated in cardiomyocytes derived from HLHS patients and in engineered cardiac models, further reinforcing the

Table 3. LncRNAs in TOF

LncRNA name	Function/mechanism (simplified)	Expression site/stage	Model system	Reference
<i>NRON</i>	Supports cardiomyocyte differentiation through signaling	Ventricular myocardium of TOF hearts	Human TOF heart tissue	70
<i>FGD5-AS1</i>	Regulates cardiac development genes via RNA interaction	Ventricular myocardium of TOF hearts (postnatal)	Human TOF heart tissue	73
<i>HA117</i>	Acts as a biomarker associated with TOF subtypes and outcomes	Right ventricle outflow tract tissue of TOF hearts	Human TOF heart tissue	74
<i>TBX5-AS1:2</i>	Regulates <i>TBX5</i> and affects cardiomyocyte proliferation	Fetal heart tissue in TOF	Human fetal heart samples	75

LncRNA indicates long noncoding RNA; and TOF, tetralogy of Fallot.

central regulatory role of lncRNAs in cardiac development and disease.^{27,46,79}

Collectively, as summarized in Table 4, current evidence indicates that (1) developmental hub lncRNAs, such as *Uph* and *Fendrr*, influence early cardiac tube and outflow tract formation through transcriptional and epigenetic regulation mechanisms; (2) the valve-specific regulatory network of *GATA6-AS1/Moshe* plays a causal role in the morphogenesis of the aortic valve and related malformations; (3) endocardial and endothelial lineage abnormalities in HLHS are closely associated with lncRNA–NOTCH/endothelial-to-mesenchymal transition network dysregulation; (4) patient-derived models have successfully recapitulated lncRNA-driven transcriptional and functional imbalances, laying the groundwork for future biomarker discovery and targeted therapeutic strategies.

LncRNAs Heterogeneity Across CHD Subtypes

At the functional level, lncRNAs exhibit remarkable heterogeneity across different CHD subtypes. In VSD and ASD, most studies have demonstrated lncRNA-mediated regulation of cardiomyocyte differentiation, cell adhesion, and structural gene expression. In contrast, TOF-associated lncRNAs, such as *FGD5-AS1*, *NRON*, and *HA117* (Table 3), are primarily involved in right ventricular remodeling and metabolic reprogramming. Studies of HLHS and transposition of the great arteries have revealed that lncRNAs participate in endothelial-to-mesenchymal transition and endocardial lineage development. Furthermore, in aortic valve and endocardial cushion malformations, *GATA6-AS1/Moshe* have been shown to directly regulate valvulogenesis.^{46,78}

These findings suggest that, although lncRNAs function within a common cardiac developmental framework, their functional consequences differ according to the specific cardiac lineage and developmental stage involved in each CHD subtype. In this context, distinct CHD phenotypes may reflect lineage-specific susceptibility to upstream lncRNA dysregulation.⁸⁰

LncRNAs Homogeneity Across CHD Subtypes

In contrast to the marked anatomic and phenotypic diversity of CHD, accumulating evidence demonstrates

remarkable consistency in lncRNA-mediated regulatory mechanisms across different CHD subtypes. Rather than functioning as disease-specific transcripts, dysregulated lncRNAs converge on a core developmental regulatory network governing cardiac morphogenesis.^{80–82}

At the signaling level, lncRNAs associated with VSD, ASD, TOF, HLHS, and valvular malformations consistently converge on canonical cardiac developmental pathways, including Wnt, TGF-β, and Notch signaling.^{80,81} Mechanistically, these lncRNAs exert conserved functions through ceRNA-mediated networks, PRC2-dependent chromatin remodeling, DNA methylation, and enhancer RNA-mediated transcriptional regulation, thereby modulating the activity of key cardiac transcription factors, such as NKX2-5, GATA4/6, TBX5, and HAND2.⁸¹

Importantly, several lncRNAs have been detected in maternal plasma and amniotic fluid-derived exosomes, suggesting their involvement in placenta-heart developmental signaling.⁸¹ This shared feature may partly explain the close association between septal defects (VSD, ASD, and TOF) and early maternal–fetal signaling imbalance (Figure 3).

Taken together, the apparent heterogeneity of CHD phenotypes conceals an underlying homogeneity in lncRNA-mediated developmental regulation. Distinct anatomic malformations may therefore arise from positional imbalances within a common lncRNA-governed cardiac developmental network, rather than from disease-specific noncoding transcripts.⁸²

Clinical Perspectives and Translational Paradigm of lncRNAs in CHD

Building on the mechanistic insights gained from different CHD phenotypes, research focus on lncRNAs is expanding from functional characterization of lncRNAs toward exploring their clinical potential in diagnosis, risk prediction, and therapeutic intervention. Owing to their high tissue specificity, molecular stability, and multilayered regulatory properties, lncRNAs are increasingly recognized as critical molecular bridges linking basic research to clinical practice.

In the context of biomarker discovery, the expression profiles of lncRNAs can sensitively reflect the status

Table 4. LncRNAs in Complex CHD

LncRNA name	Function/mechanism	Expression site/stage	Model system	Reference
FENDRR	Modifies chromatin to regulate heart and body wall development	Embryonic mesoderm (heart/body wall)	Mouse (in vivo)	28
Upperhand (Uph)	Regulates Hand2 gene expression for right ventricular and outflow tract development	Embryonic heart (right ventricle and outflow tract)	Mouse (in vivo)	29
GATA6-AS1 (Moshe)	Controls aortic valve development through signaling pathway regulation	Embryonic heart (aortic valve region)	Mouse (in vivo)	78

CHD indicates congenital heart disease; and lncRNA, long noncoding RNA.

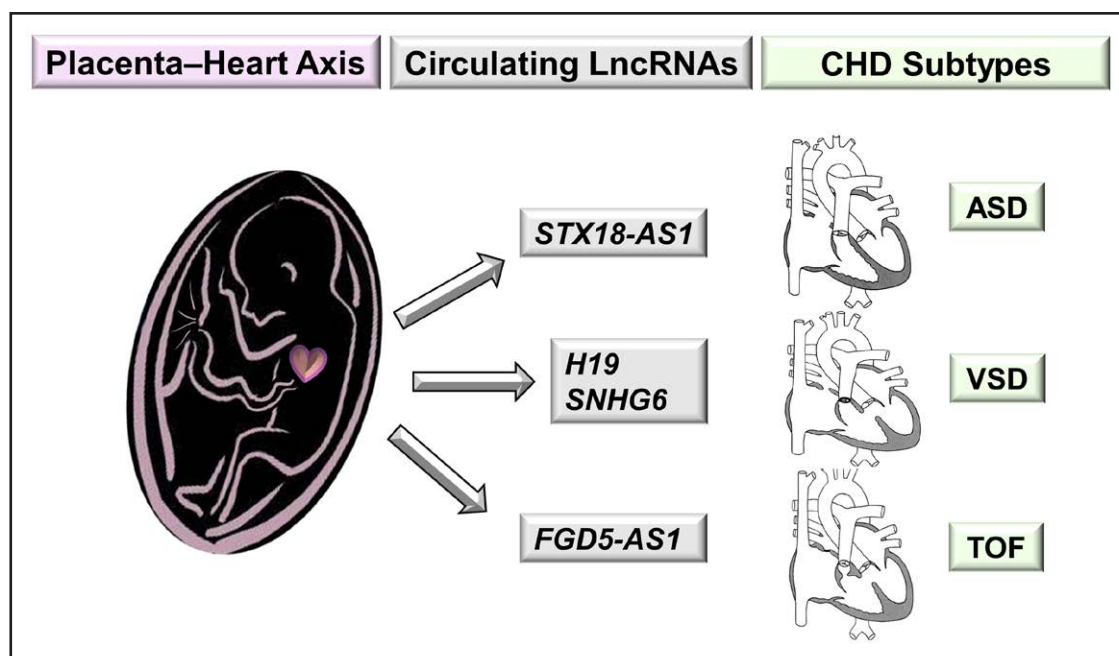


Figure 3. Placenta-heart axis and circulating long noncoding RNAs (lncRNAs) in the context of congenital heart disease (CHD). Schematic representation depicts examples of circulating lncRNAs via the placenta-heart axis and associated CHD phenotypes. ASD indicates atrial septal defect; TOF: tetralogy of Fallot; and VSD, ventricular septal defect.



of cardiac development and pathological remodeling. By modulating developmental signaling pathways and regulating miRNA-mRNA interactions through ceRNA networks, lncRNAs play essential roles in cardiomyocyte lineage differentiation and signal transduction.^{83,84}

From a genetic diagnosis perspective, integrative multiomics analyses have revealed that lncRNAs located in CNVs may serve as molecular links between structural genomic abnormalities and molecular regulatory networks, providing new avenues for molecular subtyping and pathological prediction in CHD.⁸² Meanwhile, circulating lncRNAs—such as *H19*, *MALAT1*, and *FGD5-AS1*—detected in maternal plasma, amniotic fluid, and exosomes—have demonstrated high sensitivity and specificity in the early screening of fetal cardiac defects, highlighting the feasibility of developing prenatal molecular diagnostic approaches.⁸⁴ Further studies have shown that polymorphisms in *MALAT1* gene are closely associated with CHD susceptibility in certain Chinese populations, suggesting that genetic variations in lncRNAs may influence the risk of cardiac developmental abnormalities.⁸⁵

In terms of management strategies, lncRNAs may serve not only as molecular markers of disease states but also as direct therapeutic targets. In studies of cardiac remodeling, *Wisper* (*Wisp2* super-enhancer-associated RNA) has been identified as a fibroblast-specific key lncRNA that regulates collagen synthesis and cardiac fibrosis. Importantly, antisense oligonucleotide-based inhibition of *Wisper* significantly improved cardiac function in animal models, highlighting its potential as a novel

therapeutic target.⁴⁷ In addition, other lncRNAs participate in the regulation of cardiac fibrosis through multiple signaling pathways, providing new targets for the development of anti-fibrotic nucleic acid therapies.⁸⁶ On the other hand, the *UCA1-miR-16-5p-ABCB1* axis promotes the drug transporter P-glycoprotein (P-gp) in placental models, suggesting a potential role of lncRNAs in pregnancy-related pharmacokinetics and fetal cardiac development.⁸⁷

Overall, the current spectrum of lncRNA research is widely expanding, bridging fundamental discovery to clinical application. The unique stability, detectability, and therapeutic manipulability position lncRNAs as promising molecular biomarkers and therapeutic targets for CHD. With the continued advancement of nucleic acid-based therapeutics and integrative multiomics technologies, an emerging paradigm of lncRNA-mediated precision diagnostic and therapeutic strategies is expected to facilitate early screening, risk assessment, and individualized treatment of CHD.

LIMITATIONS AND FUTURE DIRECTIONS

Methodology and Reproducibility

Technical Limitations

Current studies on lncRNAs in CHD still face significant technical and methodological limitations. Most investigations remain confined to differential expression profiling and correlative analyses, lacking functional validation and causal evidence. Moreover, heterogeneity in sample sources, RNA extraction methods, library preparation protocols, and bioinformatic

pipelines has substantially limited the reproducibility and cross-center comparability of results.⁸⁰ On the other hand, lncRNAs exhibit low sequence conservation and strong spatiotemporal regulation, making it difficult to fully extrapolate findings from animal models to human cardiac development and disease.^{26,79} Future studies should integrate human-derived systems with spatiotemporal transcriptomic technologies to enhance biological interpretability.

Cross-Species Validation

Most representative functional studies on cardiac lncRNAs, such as *Uph*, *Fendrr*, and *Bvht*, to date have been derived from animal or in vitro models. However, their dosage effects, functional redundancy, and developmental plasticity in humans remain largely unclear.^{28,29} Human single-cell omics studies—such as those investigating endocardial lineage defects in HLHS—have begun to uncover key developmental checkpoints,⁴⁶ however, limited sample sizes and incomplete temporal coverage, along with low lncRNA abundance currently restrict the transition of these findings from mechanistic insights to clinical translation.

Diagnostic Interpretation

Integration of genetic structural variants and noncoding mechanisms remains limited. Recent genomic studies have shown that CNVs within noncoding regions, coupled with lncRNA regulatory networks, may play critical roles in CHD. However, most conclusions remain predictive or correlative, lacking functional validation and quantitative assessments of causality and dosage effects.⁸² Establishing causal links among CNVs, lncRNA regulation, and phenotypic manifestations represents a key direction for future research.

Existing Gap Between Biomarker Discovery and Clinical Implementation

Circulating lncRNAs detected in amniotic fluid biopsies, such as maternal plasma, amniotic fluid, and exosomes, have shown great potential to serve as biomarkers for prenatal screening of CHD. However, significant challenges remain in cross-center standardization, assay threshold harmonization, cohort size, and real-world validation.⁷⁶ Building large-scale, multicenter prospective cohorts and standardized detection pipelines is urgently needed to facilitate the translation of lncRNA biomarkers from research to clinical practice.

FUTURE RESEARCH DIRECTIONS

High Spatiotemporal-Resolution Human Atlases

Future studies should integrate single-cell and spatial transcriptomics to construct comprehensive lncRNA–cell lineage–signaling pathway maps across key stages of human cardiac development. Functional validation can then be achieved using iPSC-derived cardiac models and organoids, providing mechanistic insights into how lncRNAs orchestrate lineage specification and morphogenesis in CHD.^{26,46} Importantly,

recent advances in hiPSC-derived cardiomyocytes have enabled patient-specific lncRNA profiling in individuals with CHD, including those with TOF, HLHS, and inherited arrhythmia syndromes such as long-QT syndrome. These studies have begun to establish links between lncRNA dysregulation and disease-specific structural and functional cardiac abnormalities in a human cellular context.⁸⁸ In the future, greater integration of patient-derived hiPSC models with cardiac organoid technologies is expected to further enhance the study of lncRNA dysregulation in CHD. Such approaches will enable more precise modeling of disease heterogeneity, facilitate functional validation of candidate lncRNAs, and accelerate the translation of lncRNA-based biomarkers and therapeutic strategies into clinical applications.⁸⁹

Long-Read RNA Sequencing for Comprehensive lncRNA Discovery

Recent advances in long-read RNA sequencing technologies enable full-length transcript reconstruction, allowing more accurate identification of transcript isoforms and previously unannotated lncRNAs. Given the structural complexity and tissue-specific expression of lncRNAs, long-read sequencing provides a distinct advantage over short-read approaches in capturing transcript diversity. In the context of CHD, this strategy may help uncover developmentally restricted or disease-specific lncRNA isoforms that are missed by conventional transcriptomic approaches, thereby refining molecular classification and mechanistic interpretation of cardiac malformations, and potentially improving early diagnosis and risk stratification.⁹⁰

Functional validation and causal inference of lncRNAs

Future research should develop high-throughput functional screening platforms based on CRISPRi/a (CRISPR interference/activation) and antisense oligonucleotide technologies to enable systematic validation of candidate lncRNAs and clarify their causal roles in CHD pathogenesis.^{20,51,80}

Integration of Noncoding Structural Variants

Future research should integrate CNV-associated lncRNAs with cis- and trans-regulatory networks, enhancer landscapes, and topological genome organization to quantitatively assess their effects on developmental thresholds and plasticity. Such integrative analyses will help elucidate how noncoding variants reshape gene regulation and contribute to congenital heart malformations.^{51,79,82}

Clinical Translation of lncRNA Biomarkers

Future efforts should focus on establishing standardized cross-platform preprocessing and quantification pipelines to ensure data comparability and reproducibility. Validation of circulating lncRNA biomarkers in prospective multicenter cohorts is needed to evaluate their sensitivity, specificity, and additive predictive value, ideally through integrated modeling with imaging and genetic diagnostic data. These steps will be essential to translate lncRNA signatures from exploratory biomarkers into robust clinical tools for CHD screening and prognosis.⁷⁶

Ethical and Regulatory Considerations

For studies involving embryonic, fetal, and perinatal samples, future research should strengthen ethical oversight, informed consent procedures, and data-sharing frameworks to ensure transparency, reproducibility, and responsible use of sensitive biological materials and multi-omics data sets. Long-term safety monitoring and regulatory standards will also be essential before lncRNA-based diagnostic or therapeutic strategies can be translated into prenatal or pediatric cardiovascular practice.

CONCLUSIONS

The development of CHD is a multifactorial and multi-stage process involving complex networks of transcriptional, epigenetic, and posttranscriptional regulation. Literature evidence demonstrates that lncRNAs are essential components of gene expression regulation that play pivotal roles in cardiac development and structural remodeling. Distinct CHD subtypes exhibit unique lncRNA expression profiles, which act in concert with classical developmental signaling pathways and RNA-binding protein networks to coordinate cardiomyocyte differentiation, migration, and functional maturation. The current landscape of lncRNA research is bridging fundamental discovery to clinical application. Specific lncRNAs not only hold great potential for early diagnostic and prognostic evaluation but also provide new avenues for targeted interventions as promising molecular biomarkers and therapeutic targets for CHD. The discovery of CNV-lncRNAs, circulating lncRNAs, and exosomal lncRNAs has expanded the translational prospects of genetic diagnosis and fluid biopsy in CHD. However, the field still faces significant limitations, including small cohort sizes, lack of standardized detection and functional validation methods, and incomplete understanding of cross-species divergence and spatiotemporal regulatory mechanisms. In the future, the rapid advance of single-cell and spatial transcriptomics, together with CRISPR-based functional editing and prospective multicenter clinical studies, will help elucidate the systemic roles and dynamic regulatory

networks of lncRNAs in the pathogenesis of CHD. By integrating variant-lncRNA-mRNA interaction modeling and epigenetic modification analyses, lncRNAs are expected to serve as critical molecular bridges linking mechanistic research to precision diagnosis and therapy for infants with CHD.

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Disclosures

None.



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